$$x - \frac{3}{2} = \pm\frac{\sqrt{17}}{2} \qquad \text{(Taking the square root on both sides)}$$

$$x = \frac{3}{2} + \frac{\sqrt{17}}{2} \text{ or } x = \frac{3}{2} - \frac{\sqrt{17}}{2}$$

Therefore, $x = \frac{3+\sqrt{17}}{2}, \frac{3-\sqrt{17}}{2}$

Example 3.31 Solve $2x^2 - x - 1 = 0$

Solution $2x^2 - x - 1 = 0$

$$x^2 - \frac{x}{2} - \frac{1}{2} = 0 \qquad (\div 2 \text{ make co-efficient of } x^2 \text{ as } 1)$$

$$x^2 - \frac{x}{2} = \frac{1}{2}$$

$$x^2 - \frac{x}{2} + \left(\frac{1}{4}\right)^2 = \frac{1}{2} + \left(\frac{1}{4}\right)^2$$

$$\left(x - \frac{1}{4}\right)^2 = \frac{9}{16} = \left(\frac{3}{4}\right)^2$$

$$x - \frac{1}{4} = \pm\frac{3}{4} \Rightarrow x = 1, -\frac{1}{2}$$

Solving a Quadratic Equation by Formula Method

The formula for finding roots of a quadratic equation $ax^2 + bx + c = 0$ (derivation given in section 3.6.2) is $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.



The formula for finding roots of a quadratic equation was known to Ancient Babylonians, though not in a form as we derived. They found the roots by creating the steps as a verse, which is a common practice at their times. Babylonians used quadratic equations for deciding to choose the dimensions of their land for agriculture.

Example 3.32 Solve $x^2 + 2x - 2 = 0$ by formula method

Solution Compare $x^2 + 2x - 2 = 0$ with the standard form $ax^2 + bx + c = 0$

$a = 1, b = 2, c = -2$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

substituting the values of a, b and c in the formula we get,

$$x = \frac{-2 \pm \sqrt{(2)^2 - 4(1)(-2)}}{2(1)} = \frac{-2 \pm \sqrt{12}}{2} = -1 \pm \sqrt{3}$$

Therefore, $x = -1 + \sqrt{3}, -1 - \sqrt{3}$

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